

Mason Bane

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Summary: Computer Science student (Computer Engineering) targeting embedded firmware and hardware/software integration roles. Proficient in microcontroller programming, low-level C/C++ and ARM Assembly, and circuit design. Strong mathematical/analytical skills with research experience.

EDUCATION

Tarleton State University

Bachelor of Science in Computer Science, Concentration in Computer Engineering

Minor in Mathematics

GPA: 3.94/4.00 (Institutional) — Cumulative: 3.75/4.00

Stephenville, TX

Aug. 2022 – May 2026

Hill College

Associate of Arts in Liberal Arts

Hillsboro, TX

Aug. 2019 – Sept. 2023

EXPERIENCE

Undergraduate Research Assistant - Lead Programmer

May 2024 – May 2026

Tarleton State University

Stephenville, TX

- Developed and optimized various N-Body simulations utilizing C/C++, enhancing computational efficiency and accuracy through advanced algorithms and parallel processing with CUDA.
- Collaborated with interdisciplinary teams to create digital twins to model complex problems using OpenGL and Blender, translating scientific concepts into interactive simulations and improving data interpretation.
- Conducted rigorous testing and debugging of simulation software, ensuring robust performance and reliability while documenting processes to facilitate knowledge transfer and future research initiatives.

Undergraduate Technology Specialist - HPC Lab Manager

Aug. 2024 – May 2026

Tarleton State University

Stephenville, TX

- Managed and maintained 15 Linux devices in a high-performance computer lab dedicated to research initiatives
- Performed system updates, hardware maintenance, and technical support, resolving issues promptly to minimize downtime and maintain optimal performance.
- Maintained a clean and organized lab environment, promoting a collaborative workspace that fosters innovation and productivity.

PROJECTS

Secure IoT Sensor Node with Hardware TrustZone | C, STM32CubeIDE/HAL, I2C/SPI Dec. 2025 – May 2026

- Implementing a secure data pipeline on an STM32H5 MCU, utilizing Arm TrustZone and the Secure Manager to create a hardware-rooted chain of trust for environmental sensor data.
- Developing drivers and application logic to interface with a Bosch BME280 temperature, humidity, and pressure sensor via I2C communication protocol.
- Structuring firmware to cryptographically sign sensor readings within the secure processing environment for tamper-evident logging.
- Utilizing STM32CubeIDE and the HAL for peripheral configuration and hardware security module (HSM) management as a foundation for secure telemetry systems.

N-body Digital Twin of the Left Atrium | NIH Grant #1R15HL179671-01 | C, CUDA Aug. 2024 – May 2026

- Engineered a parallel N-body model of the left atrium using CUDA, with a 20,000+ node mesh to simulate atrial arrhythmias in near real-time.
- Developed an intuitive C++/ImGui interface to control simulation parameters and visualize outputs, bridging the gap between complex computational models and end-user (research/clinical) requirements.
- Presented findings at academic conferences, demonstrating the tool's potential to improve clinical decision-making and transform training for medical professionals.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, ARM Assembly (Cortex-M), Java, Bash

Microcontroller Platforms: STM32 (H5/Cortex-M33), TIVA-C (TM4C), Arduino, Raspberry Pi

Embedded Protocols & Tools: I²C, SPI, UART, GPIO, STM32CubeIDE, GDB, RTOS Concepts

Systems & Development: Git/GitHub, Linux, CMake/Make, Visual Studio, VS Code

Simulation & Design: LTSpice, MATLAB, CUDA, OpenGL, ImGui, Blender